SECTION

LINEARITY IN CALIBRATION

6

CHAPTER

28 LINEARITY IN CALIBRATION—ACT I[☆]

Those who know us know that we have always been proponents of the approach to calibration that uses a small number of selected wavelengths. The reasons for this are partly historical, since we became involved in chemometrics through our involvement in NIR, back when individual-wavelength-based calibration techniques (what we now call "MLR") were essentially the only ones available, and these methods did yeoman's service for many years. When full-spectrum methods (PCR (principal component regression), PLS (partial least squares)) came on the scene and became popular, we adopted them as another set of tools in our chemometric armamentarium, but always kept in mind our roots, and use individual-wavelength-based techniques when necessary and appropriate. We always knew that they could sometimes perform better than the full-spectrum techniques under the proper conditions, despite all the hype of the proponents of the full-spectrum methods. Lately, various other workers using full-spectrum calibration algorithms (i.e., PCR and PLS) have also noticed that eliminating "extra" wavelengths could improve the results, but nobody (including ourselves) could predict when this would happen or explain or define the conditions that make it possible. The advantages of the full-spectrum methods are obvious and are promoted by the proponents of the full-spectrum methods at every opportunity: the ability to reduce spectral noise by averaging data over both wavelengths and spectra, rejecting noise by rejecting the higher factors, into which noise is preferentially placed, the advantages inherent in the use of orthogonal variables, and the avoidance of the time-consuming step of performing the wavelength-selection process.

The main problem was to define the conditions where wavelength selection was superior; we could never quite put our finger on what characteristics of spectra would allow the wavelength-based techniques to perform better than the full-spectrum methods.

Until recently

What sparked our realization of (at least one of) the key characteristics was an online discussion of the NIR discussion group[1] dealing with a similar question, whereupon the ideas floating around in our heads congealed. At the time, the concept was proposed simply as a thought experiment, but afterward

[☆]Adapted from Spectroscopy 13(6) (1998) 19–21.

[☆☆]Postscript (We retain this postscript for historical reasons, although after 20 years we can no longer rely on the validity of the information.): After the first edition of this book was published, the sponsorship and support of the discussion group described above was changed. It is currently sponsored and supported by IM Publications. Inquiries can be sent to ian@impublications.com.

[1](We retain this footnote for historical reasons, although after 20 years we can no longer rely on the validity of the information.): The moderator of this discussion group was Bruce Campbell. He can be reached for information or to join the discussion group by sending a message to CAMPCLAN@prodigy.net. New members are welcome.

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CHAPTER

LINEARITY IN CALIBRATION—ACT II SCENE I

29

When the preceding chapter "Linearity in Calibration" (Chapter 28) was first published as a Spectroscopy column [1] we didn't quite realize what a firestorm we were going to ignite, although, truth be told, we did not expect everybody to agree with us, either. But if so many actually took the trouble to send their criticisms to us, then there must also be a large "silent majority" out there that are upset, perhaps angry, and almost certainly misunderstanding what we said. We prepared responses to these criticisms, but they became so lengthy that we could not print them all in a single column or chapter; therefore, they are spread out over the current chapter and the next few. We will include the responses we received at an appropriate point, followed by our comments about them.

Before doing that, however, let's discuss the question of the linearity of spectroscopic data in general, first. There are a number of causes of nonlinearity that most chemists and spectroscopists are familiar with. Let us define our terms. When speaking of "linearity" the meaning of the term depends on your point of view and your interests. An engineer is concerned, perhaps, with the linearity of detector response as a function of incident radiant energy. To a chemist or spectroscopist, the interest is in the linearity of an instrument's response as a function of the concentration of an analyte in a set of samples. In practice, this is generally interpreted to mean that when measuring a transparent, nonscattering sample, the response of the instrument can be calculated as some constant times the concentration of the analyte (or at least some function of the instrument response can be calculated as a constant times some other function of the concentration). In spectroscopic usage, that is normally interpreted as meaning the condition described theoretically by Beer's law, that is, the instrument response function is the negative exponential of the concentration:

$$I = k I_o e^{-bC} \tag{29-1}$$

where

I = the radiation passing through the sample.
k = the multiplying constant.
I_o = the radiation incident on the sample.
b = the product of the pathlength and absorbtivity.
C = the concentration of the analyte.

When other types of samples are measured, the resulting data is usually known to be nonlinear (with respect to the analyte concentration, except possibly in a few special cases) so those measurements are of no interest to us here. Thus, in practice, the invocation of the term "linearity" implies the assumption that Beer's law holds; therefore, discussions of nonlinearity are essentially about those phenomena that cause departures from Beer's law.

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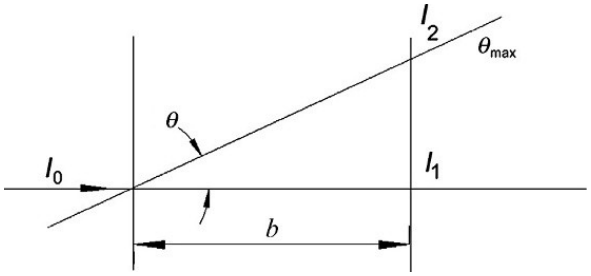


FIG. 29-1

A ray striking the sample at an angle Θ to the Normal of the surface (I_2) traverses a greater distance through the sample than one that strikes at the Normal angle (I_1).

that there is a fundamental limitation. Mechanistically, the nonlinearity is caused by the fact that the absorbance for the rays traveling normally $= abc$, while for the extreme rays, it is $abc/\cos(\theta_{max})$. Thus, the nonnormal rays pass through a longer pathlength within the sample and as a consequence suffer higher absorbance than the normal ones do, and the discrepancy (which equals $abc(1–1/\cos(\theta))$) increases with increasing concentration.

When the medium is completely nonabsorbing then the difference in pathlength does not affect the measurement. When the sample has absorbance, however, it is clear that ray I_2 will have its intensity reduced more than ray I_1, due to the longer pathlength. Thus all rays are reduced by a different amount and this leads to the nonlinearity of the measurement. Mathematically, this can be expressed by noting that the intensity measured when a beam with a finite range of angles passes through a sample is

$$I = I_o \int_0^{\theta_{max}} e^{-b/\cos(\theta)} d\theta \tag{29-5}$$

rather than the simpler form shown in Eq. (29-1) (which, we remind the reader, only holds true for "perfectly collimated" beams, which have zero energy).

In practice, of course, this effect is very small, normally much smaller than any of the other sources of nonlinear behavior, and we are ordinarily safe in ignoring it and calling Beer's law behavior "linear" in the absence of any of the other known sources of nonlinear behavior. However, the point here is that this completes the demonstration of our statement above, that Beer's law never exactly holds EVEN IN PRINCIPLE and that as spectroscopists we never ever really work with perfectly linear data.

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CHAPTER

30 LINEARITY IN CALIBRATION—ACT II SCENE II: READER RESPONSES☆

In a previous chapter (Chapter 28), entitled "Linearity in Calibration," based on the Spectroscopy column [1], we presented some unexpected results when comparing a calibration model using MLR with the model found using PCR. That Spectroscopy column generated an active response, so we are discussing the subject in some detail, spread over the next few chapters.

In this chapter, we now present the responses we received to the original column, following which we will comment about them in subsequent chapters. Here, in order of receipt, are the comments.

The first set of comments we received were from Richard Kramer:

[Howard & Jerry],

I'm afraid that this month's Spectroscopy Column is badly off the mark (pun intended (with apologies)). The errors are twofold with the most serious error so significant that the other error is moot.

1. If I understand the column correctly, a 1-factor model was used. Well, a single linear factor can never be sufficient to properly model a nonlinear system. A minimum of 2 factors are required. The synthetic data did NOT demonstrate the advantage of a single linear wavelength over a multiple wavelength model, it merely illustrated the fact that a single linear factor is not sufficient to model nonlinear data. We could stop here, but, for the sake of completeness.
2. The second problem is that that we never have the luxury of working with noise-free data. Thus, the column did not ask the right question(s). The proper question to ask is "In what ways and under which circumstances do the signal averaging advantages of multiple-wavelength models outperform or underperform with respect to a single waveelngth (or n wavelengths, where *n* is a small integer) calibration when noise is present?" The answer will depend upon the levels of noise and nonlinearity and the number of wavelengths in each model.

Regards, Richard.

We went back and forth a couple of times but rather than list each of our conversations individually, we will reserve our comments until we have looked at all the reader's comments, and then we will summarize our responses to all four respondents together, since several of these response comments say the same things, to some extent.

Second, we received comments from Patrick Wiegand.

Gents,

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To summarize the comments (including ones presented during subsequent discussions and therefore not included above):

1. Richard Kramer, Patrick Wiegand, and Fred Cahn felt that we should have tried two factors.
2. Richard Kramer and Patrick Wiegand thought we should have added simulated noise to the data.
3. All four responders indicated that we should have tried PLS.
4. Richard Kramer, Patrick Wiegand, and Paul Chabot indicated that one PLS factor might do as well as one wavelength.
5. Richard Kramer and Patrick Wiegand thought that our conclusion was that MLR is better than PCA.

As stated in the introduction to this chapter, we present our responses in a subsequent chapter.

REFERENCE

[1] H. Mark, J. Workman, Chemometrics in spectroscopy - part 27 - linearity in calibration, Spectroscopy 13 (6) (1998) 19–21.

CHAPTER

LINEARITY IN CALIBRATION—ACT II SCENE III[☆] 31

In Chapter 28, titled "Linearity in Calibration," which itself was based on a Spectroscopy column [1], we presented some unexpected results when comparing a calibration model using MLR with the model found using PCR. That column generated a rather active response (reproduced in Chapter 30), so we are discussing the subject and responding to the comments received in some detail, spread over several chapters. We presented the list of reader's comments in Chapter 30, which began and ended with a summary of the comments received regarding the "Linearity in Calibration" column. We therefore pick up where we left off by starting this chapter with that same summary (naturally, anyone who wishes to read the full text of the comments will have to go back and reread Chapter 29):

1. Richard Kramer, Patrick Wiegand, and Fred Cahn felt that we should have tried two factors.
2. Richard Kramer and Patrick Wiegand thought we should have added simulated noise to the data.
3. All four responders indicated that we should have tried PLS.
4. Richard Kramer, Patrick Wiegand, and Paul Chabot indicated that one PLS factor might do as well as one wavelength.
5. Richard Kramer and Patrick Wiegand thought that our conclusion was that the MLR is better than PCA.

In addition, each of the responders had some of their own individual comments; we discuss all these below.

We now continue with our discussion of these comments: It may surprise some to hear this, especially in the light of some of the comments we make below, but we agree with the responders more than we disagree. We also believe, for example, in prescreening the data, at least as strongly as Patrick Wiegand does, and we believe his comments regarding the way all (or at least, let's hope all) experienced chemometricians approach a problem. Indeed, fully half the book that one of us authored [2] was spent on just that point: How to "look at the data." However, our experience in the "real world" (as some like to call it) of instrument manufacturers has given us a somewhat different slant on the reality of what actually happens when users get hold of a new super-whiz-bang package of calculation.

In 16 years of experience in the NIR Applications Department at Technicon Instruments, there was about an hour and a half available to teach both theory and practice of calibration to each group of new users; the rest of the training time was spent teaching the students how to set the instrument up, prepare samples, take reproducible readings, and learn the rest of the mechanics needed to run the instrument, take readings, and collect the data. How much attention do you think could be paid to the

[☆]Adapted from Spectroscopy 14(2) (1999) 16–27, 80.

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improved the results (in PCR and PLS as well as MLR), but there has previously been no explanation for this phenomenon. Therefore, we decided to investigate nonlinearity since we suspected that to be a major consideration and so it turned out to be.

We continue our discussion in the next chapter.

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CHAPTER

LINEARITY IN CALIBRATION—ACT II SCENE IV☆

32

We continue our discussion started by the responses received to our Spectroscopy column "Linearity in Calibration" (reproduced in Chapter 30). So far our discussion has extended over three Chapters 29–31.

In Chapter 31, we stated: "we are not fighting a 'holy war' against PCA/PLS etc." and then went on to discuss what our original column was really about.

However, if there is a "holy war" being fought at all, then from our point of view, it is against the practice of simply accepting the results of the computer's cogitations without attempting to understand the underlying phenomena that affect the behavior of the calibration models, regardless of the algorithm used. This has been our fight since the beginning—which can be verified by going back and rereading our very first column ever [1] (reproduced in our book [2]).

Jerry and I don't always agree, but we do agree on the following: It is incomprehensible to us how a person calling himself a scientist can fail to wonder WHY calibration models behave the way they do and try to relate their behavior to the properties of the data giving rise to them. There are reasons for everything that happens, whether we know what those reasons are or not, and the goal of science is to determine what those underlying reasons or principles are. At least that is the goal of every other field of scientific endeavor that we are aware of—why is chemometrics exempt?

Real data, as we have seen, is far too complicated to work with to try to obtain fundamental understanding, just as the physical world is often too complicated to study directly in toto. Therefore work such as was presented in Chapter 29 is needed, creating a simplified system where the characteristics of interest can be isolated and studied—just as physical experiments often work with a simplified portion of the physical world for the same reason. This might be categorized as "Experimental Chemometrics": Controlling the nature of the data in a way that allows us to relate the properties of the data to the behavior of the model. Does this mimic the "real world?" No, but it does provide a window into the inner workings of the calibration calculations, and we need as many such windows as we can get.

We will go so far as to make an analogy with Chemistry itself. The alchemists of old had an enormous empirical knowledge base and from that could do all manner of useful things. But we do not now consider alchemy a science, and it did not become the science we now call chemistry until the underlying principles and phenomena were discovered and codified in a way that all could use. The current state of chemometrics is more nearly akin to alchemy than chemistry: We can do all manner of useful things with it, but it is all empirical and there are still many areas where even the most expert and prominent practitioners treat it as a "black box" and make no attempt to understand the inner workings of that black box. Empiricism is important and even necessary but hardly sufficient. The ultimate test of

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come by, this loss is not too serious, but what do you do when a project requires calibration runs that cost a million (or 10 million) dollars per run, and minimizing the number of runs is the absolute top priority? In such a case, you will not only not have validation data, you will likely not even have enough calibration data to do a leave-one-out calculation, and then being able to evaluate models from calibration diagnostics alone will be critical. Statisticians have, in fact, developed diagnostic tests that provide information about such characteristics, but the chemometric community, in our arrogance, thinks we know better and ignores all this prior work. The statistical community has also developed many local and semilocal diagnostic tools to help understand and improve calibration models; we really need to get back to the roots on this, as well.

There are innumerable unsolved problems in chemometrics that need to be addressed: real, scientific problems, not just new ways to throw numbers around. Anyone who needs help in thinking of some can check out Chapter 27 where we presented a few; maybe those will jog the innovative juices.

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CHAPTER

LINEARITY IN CALIBRATION—ACT II SCENE V☆

33

This chapter is the further continuation of our discussion started by the responses received to Chapter 29 "Linearity in Calibration" (when published as a column in Spectroscopy). So far our discussion has extended over three previous chapters. At this point, however, we are finally getting toward the end of our obsession with considerations of linearity—at least until we receive another set of comments from our readers. Incidentally, we welcome such feedback, even those that disagree with us or with which we disagree, so please keep it coming. Indeed, it seems that we don't get much feedback unless our readers disagree with us and feel it strongly enough to have the need to say so. That's great—there's nothing like a little controversy to keep a discussion like this interesting: Who said chemometrics and statistics and mathematics were dry subjects, anyway?!

In our first chapter on this topic (Chapter 29) we had only done a PCA to compare with the MLR results. One of the comments made, and it was made by all the responders, was to ask why we didn't also do a PLS analysis of the synthetic linearity data. There were a number of reasons, and we offered to send the data to any or all of the responders who would care to do the PLS analysis and report the results. Of the original responders, Paul Chabot took us up on our offer. In addition, at the 1998 International Diffuse Reflectance Conference (The "Chambersburg" meeting), Susan Foulk also offered to do the PLS analysis of this data.

Gratifyingly, when Paul and Susan reported their PLS loadings they were identical, even though they used different software packages to do the PLS calculations (PLSIQ and Unscrambler). We're certainly glad we don't have to worry about sorting out differences in software packages (often due to different convergence criteria, etc., that sometimes creep into results such as these) on top of the chemometric issues we want to address.

Fig. 33-1 presents the plot of the PLS loadings. Paul and Susan each computed both Loadings. Note that the first Loading is indistinguishable to the eye from the first PCA Loading (see Chapter 29).

Paul and Susan each also computed the two calibration models and performance statistics for both models. Except that various programs didn't compute the same sets of performance statistics (although in one case a different computation seemed to be given the same label as SEE), the ones that were reported by both programs had identical values.

As expected by all responders, and by your hosts as well, when two-factor models (either PCR or PLS) were computed, the fit of the model to the synthetic data was perfect. Table 33-1 presents a summary of the numerical results obtained, for one-factor calibration models.

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In their day, PCs and partial least squares were considered almost as "the magic answer to all calibration problems." It took a long time for the realization to dawn that they contain no "magic" and are subject to most of the same problems as the algorithm previously available (at that time, what we now call MLR). Now we see a surge in other new algorithms: wavelets, neural networks, genetic algorithms, as well as the combining of techniques (e.g., selecting wavelengths before performing a PCA or PLS calculation). While some of the veterans of the "PC wars" (*not* "political correctness," by the way) realize that they can be overfit just as MLR calibrations can, have become wary of the problem, and are more cautious with new algorithms, there is some evidence that a large number, perhaps the majority, of users are not nearly so careful and are still looking for their "magic answer." There is a generic caution that needs to be promoted, and all users made aware of when dealing with these more sophisticated methods. That is the simple fact that every new parameter that can be introduced into a calibration procedure is another way to overfit and hide the fact that it is happening. Worse, the more sophisticated the algorithm the harder it is to see and recognize that is what is going on.

With PCR and PLS we introduced the extra parameter of the number of factors: one extra parameter. With wavelets we introduce the order and the locality of each wavelet: two extra parameters. With neural nets, we have the number of nodes in each layer: n extra parameters, and then there is even a metaparameter—the number of layers. No wonder reports of overfitting abound (and don't forget: Those are only the ones that are recognized)!

And nary a diagnostic in sight.

In a perfect world, a new algorithm would not be introduced until a corresponding set of diagnostic methods were developed to inform the user how the algorithm was behaving. As long as we're dreaming, let's have those diagnostics be informative, in the sense that if the algorithm was misbehaving, it would point the user in the proper direction to fix it.

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